

EN.553.634 Elements of Statistical Learning

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0.1 Introduction to Statistical Learning

Let $(X, Y) \sim P$, where $X \in \mathcal{X} \subseteq \mathbb{R}^p$ is a feature vector and Y is a response. Statistical learning uses a sample $\mathcal{D}_n = \{(X_i, Y_i)\}_{i=1}^n$ to construct a prediction rule $f : \mathcal{X} \rightarrow \mathcal{A}$. In regression, $\mathcal{A} = \mathbb{R}$; in classification, $\mathcal{A}$ is a finite set or a vector of class probabilities.

Prediction and interpretation are different goals. A predictive model is judged on new observations from the deployment distribution. A coefficient can be interpreted causally only with additional identification assumptions; predictive accuracy alone does not provide them.

Example 0.1.1 (From coin flips to supervised learning). Suppose a coin's probability of heads depends on measurable conditions X , and $Y \in \{0, 1\}$ records the outcome. Estimating one bias p becomes estimating the conditional probability

$$\eta(x) = P(Y = 1 \mid X = x).$$

A classifier converts $\eta(x)$ into an action, while a calibrated probability model attempts to estimate $\eta(x)$ itself.

0.1.1 Loss Functions

A loss $L(y, a)$ quantifies the cost of predicting a when the outcome is y . Common choices are

$$\begin{aligned} L_{\text{sq}}(y, a) &= (y - a)^2, & L_{\text{abs}}(y, a) &= |y - a|, \\ L_{01}(y, a) &= \mathbb{1}\{y \neq a\}, & L_{\log}(y, q) &= -y \log q - (1 - y) \log(1 - q). \end{aligned}$$

Squared loss strongly penalizes large residuals; absolute loss is more robust. Zero-one loss targets classification accuracy but is discontinuous and hard to optimize. Logistic loss is a smooth convex surrogate and estimates class probabilities when the model is well specified.

A loss should reflect the decision. For asymmetric binary costs c_{10} for a false positive and c_{01} for a false negative,

$$L(y, a) = c_{10} \mathbb{1}\{y = 0, a = 1\} + c_{01} \mathbb{1}\{y = 1, a = 0\}.$$

0.1.2 Risk and Empirical Risk Minimization

The population and empirical risks are

$$R(f) = \mathbb{E}[L(Y, f(X))], \quad \hat{R}_n(f) = \frac{1}{n} \sum_{i=1}^n L(Y_i, f(X_i)). \quad (0.1.1)$$

Given a function class $\mathcal{F}$, empirical risk minimization (ERM) chooses $\hat{f} \in \arg \min_{f \in \mathcal{F}} \hat{R}_n(f)$. If f^* minimizes risk over all measurable rules, then

$$R(\hat{f}) - R(f^*) = \underbrace{R(\hat{f}) - \inf_{f \in \mathcal{F}} R(f)}_{\text{estimation error}} + \underbrace{\inf_{f \in \mathcal{F}} R(f) - R(f^*)}_{\text{approximation error}}.$$

The first term reflects finite data; the second reflects the restricted model class. Interpolation of training data alone gives no guarantee of generalization.

0.1.3 Optimal Prediction

Conditioning on $X = x$ reduces prediction to a pointwise decision:

$$f^*(x) \in \arg \min_a \mathbb{E}[L(Y, a) \mid X = x]. \quad (0.1.2)$$

Under squared loss, $f^*(x) = \mathbb{E}[Y \mid X = x]$; under absolute loss, any conditional median is optimal. Under zero-one loss, the optimal classifier chooses the most probable class.

Squared-loss calculation. Writing $m(x) = \mathbb{E}[Y \mid X = x]$,

$$\mathbb{E}[(Y - a)^2 \mid X = x] = \text{Var}(Y \mid X = x) + \{m(x) - a\}^2,$$

which is minimized by $a = m(x)$. □

0.1.4 The Regression Function

The regression function $m(x) = \mathbb{E}[Y \mid X = x]$ describes the systematic part of a response. Writing

$$Y = m(X) + \varepsilon, \quad \mathbb{E}[\varepsilon \mid X] = 0,$$

does not require normality or constant variance. The conditional variance $\sigma^2(x) = \text{Var}(Y \mid X = x)$ may depend on x . For Bernoulli Y , $m(x) = P(Y = 1 \mid X = x)$ is both the regression function and the class probability.

0.1.5 Generalization Error

Conditional on the training data, test error is

$$\text{Err}(\mathcal{D}_n) = \mathbb{E}[L(Y^{\text{new}}, \hat{f}(X^{\text{new}})) \mid \mathcal{D}_n].$$

Training error reuses observations that selected $\hat{f}$ and is optimistically biased. Test observations must be excluded from every modeling choice, including imputation, scaling, feature selection, and tuning.

Ordinary validation assumes that training and deployment observations have the same distribution. Covariate shift changes P_X but not $P_{Y|X}$; concept drift changes the conditional law. Random cross-validation on historical data does not diagnose either automatically.

0.1.6 The Bias-Variance Tradeoff

For squared-error prediction at fixed x , averaging over training sets gives

$$\mathbb{E}[(Y_0 - \hat{f}(x))^2 \mid X_0 = x] = \sigma^2(x) + \{\mathbb{E}[\hat{f}(x)] - m(x)\}^2 + \text{Var}(\hat{f}(x)). \quad (0.1.3)$$

These are irreducible noise, squared bias, and variance. Restrictive models tend toward higher bias and lower variance; flexible models tend toward lower approximation bias and higher sampling variance. This exact additive decomposition is specific to squared loss.

0.2 Least Squares

Given $X \in \mathbb{R}^{n \times p}$ and $y \in \mathbb{R}^n$, ordinary least squares solves

$$\hat{\beta} \in \arg \min_{\beta \in \mathbb{R}^p} \|y - X\beta\|^2. \quad (0.2.1)$$

The normal equations are $X^\top X \hat{\beta} = X^\top y$. If X has full column rank,

$$\hat{\beta} = (X^\top X)^{-1} X^\top y.$$

Computationally, solve the system using QR or SVD rather than explicitly forming the inverse; forming the Gram matrix squares the condition number.

0.2.1 Estimation and Geometry

The fitted vector is the orthogonal projection of y onto the column space:

$$\hat{y} = Hy, \quad H = X(X^\top X)^{-1} X^\top.$$

The hat matrix is symmetric and idempotent. The residual $e = y - \hat{y}$ is orthogonal to every column of X . If X is rank deficient, fitted values remain unique but coefficients do not; $X^+ y$ is the minimum-norm solution.

Leverage $h_{ii} = H_{ii}$ measures how unusual row i is in feature space. Since $\sum_i h_{ii} = \text{tr}(H) = p$, average leverage is p/n . The leave-one-out residual satisfies

$$y_i - \hat{y}_i^{(-i)} = \frac{e_i}{1 - h_{ii}}, \quad (0.2.2)$$

so OLS leave-one-out error can be computed without n refits.

0.2.2 Sampling Properties of Least Squares

Under the fixed-design model

$$y = X\beta + \varepsilon, \quad \mathbb{E}[\varepsilon \mid X] = 0, \quad \text{Cov}(\varepsilon \mid X) = \sigma^2 I,$$

OLS is conditionally unbiased and

$$\text{Cov}(\hat{\beta} \mid X) = \sigma^2 (X^\top X)^{-1}.$$

Gauss–Markov says OLS has smallest covariance among linear unbiased estimators; it does not require Gaussian errors. Normality gives exact t and F inference. An unbiased error-variance estimator is

$$\hat{\sigma}^2 = \frac{\|y - X\hat{\beta}\|^2}{n - p}.$$

With heteroskedastic errors, use sandwich standard errors or weighted least squares. If $\mathbb{E}[\varepsilon \mid X] \neq 0$, robust standard errors do not remove coefficient bias.

Example 0.2.1 (Omitted-variable bias). If $Y = \beta_0 + \beta_1 X + \beta_2 Z + \varepsilon$ but Z is omitted, the population slope from regressing Y on X is

$$\beta_1 + \beta_2 \frac{\text{Cov}(X, Z)}{\text{Var}(X)}.$$

Bias requires both a nonzero effect of Z and association between X and Z .

0.3 Multiple Regression

An intercept, numeric predictors, and encoded categorical variables are columns of a design matrix. With an intercept, a K -level categorical predictor uses $K - 1$ reference indicators. The coefficient of X_j is the change in fitted response per unit change in X_j , holding the other represented features fixed.

The Frisch–Waugh–Lovell theorem says the coefficient of X_j equals the slope obtained by residualizing both Y and X_j against the remaining predictors. Multiple regression therefore compares variation in X_j not linearly explained by the other columns.

0.3.1 Interactions and Basis Expansions

An interaction permits one effect to depend on another:

$$m(x_1, x_2) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2.$$

Then $\partial m / \partial x_1 = \beta_1 + \beta_{12} x_2$, so β_1 is the effect of x_1 only at $x_2 = 0$. Centering can improve interpretation.

A basis expansion uses fixed transformations h_m :

$$f(x) = \sum_{m=1}^M \beta_m h_m(x).$$

The model is nonlinear in x but linear in β . Raw high-degree polynomials are often unstable; orthogonal polynomials or splines are usually better conditioned.

0.4 Logistic Regression

For $Y \in \{0, 1\}$, logistic regression models

$$p(x) = P(Y = 1 \mid X = x), \quad \log \frac{p(x)}{1 - p(x)} = \beta_0 + x^\top \beta. \quad (0.4.1)$$

Thus $p(x) = \{1 + \exp[-(\beta_0 + x^\top \beta)]\}^{-1}$. A unit increase in x_j multiplies conditional odds by e^{β_j} holding other modeled features fixed; it does not add a constant amount to probability.

0.4.1 Likelihood

Including the intercept in x_i , the Bernoulli log-likelihood is

$$\ell(\beta) = \sum_{i=1}^n [y_i x_i^\top \beta - \log\{1 + e^{x_i^\top \beta}\}]. \quad (0.4.2)$$

If $p_i = p(x_i)$, then

$$\nabla \ell(\beta) = X^\top (y - p), \quad \nabla^2 \ell(\beta) = -X^\top W X,$$

where $W = \text{diag}\{p_i(1 - p_i)\}$. The likelihood is concave.

Under perfect separation, a hyperplane correctly separates all observations and the likelihood supremum occurs as $\|\beta\| \rightarrow \infty$; no finite unpenalized MLE exists. Ridge regularization gives a finite solution. Numerically, evaluate $\log(1 + e^z)$ with a stable softplus routine.

0.4.2 Fitting Logistic Regression: Newton and IRLS

Newton's update is

$$\beta^{\text{new}} = \beta^{\text{old}} + (X^\top W X)^{-1} X^\top (y - p). \quad (0.4.3)$$

With working response $z = X\beta + W^{-1}(y - p)$, this is weighted least squares:

$$\beta^{\text{new}} = (X^\top W X)^{-1} X^\top W z.$$

This is iteratively reweighted least squares (IRLS). Implementations solve the weighted system rather than invert it, and use step halving or line search if a full Newton step decreases the likelihood. Growing coefficients and fitted probabilities near zero or one suggest separation.

0.5 Bayes Classification

Let $Y \in \{1, \dots, K\}$ and $\eta_k(x) = P(Y = k \mid X = x)$. Under zero-one loss, the Bayes classifier is

$$g^*(x) = \arg \max_k \eta_k(x),$$

with Bayes error $R^* = \mathbb{E}[1 - \max_k \eta_k(X)]$.

Generative classifiers model priors π_k and class densities f_k , then apply

$$P(Y = k \mid X = x) = \frac{\pi_k f_k(x)}{\sum_j \pi_j f_j(x)}.$$

LDA assumes normal class distributions with a shared covariance, producing linear boundaries. QDA allows separate covariance matrices and quadratic boundaries, trading lower bias for substantially more parameters.

0.5.1 Thresholds

With false-positive cost c_{10} and false-negative cost c_{01} , predict class 1 when

$$\eta(x) > \frac{c_{10}}{c_{10} + c_{01}}. \quad (0.5.1)$$

The threshold $1/2$ is optimal only for equal costs. Changing a threshold changes decisions, not ranking or calibration. Select it on validation data using deployment costs or capacity constraints.

0.5.2 ROC and Calibration

For threshold t ,

$$\text{TPR}(t) = P(\hat{\eta}(X) > t \mid Y = 1), \quad \text{FPR}(t) = P(\hat{\eta}(X) > t \mid Y = 0).$$

The ROC curve plots TPR against FPR. AUC is the probability that a random positive receives a higher score than a random negative, with half credit for ties. It assesses ranking, not probability accuracy, and may hide behavior in the operational low-FPR region.

A score is calibrated when $P(Y = 1 \mid \hat{\eta}(X) = q) = q$. Reliability plots, log loss, and Brier score assess calibration. Platt scaling fits a logistic recalibration; isotonic regression fits a monotone map. Under class imbalance, precision–recall curves are often more informative because precision depends on prevalence.

0.6 Estimating Generalization Error

The quantity of interest is performance of the entire training procedure, including tuning. A test set estimates this only if it is untouched until the procedure is finalized. Repeatedly checking test performance turns the test set into training information.

0.6.1 Validation and CV

In a validation split, train on one subset, choose hyperparameters on another, and evaluate once on a test subset. In K -fold cross-validation (CV), partition indices into folds $I_1, \dots, I_K$

and compute

$$\widehat{\text{Err}}_{\text{CV}} = \frac{1}{n} \sum_{k=1}^K \sum_{i \in I_k} L\{Y_i, \hat{f}^{(-k)}(X_i)\}.$$

Five- or ten-fold CV is a common variance–computation compromise. Leave-one- out CV has nearly maximal training sets but can have high variance and is not always preferable.

Preprocessing must be refit within each training fold. For grouped, longitudinal, spatial, or time-ordered data, folds must respect dependence and deployment direction. Random row splits can leak subjects or future information.

Nested CV uses inner folds for tuning and outer folds for evaluation. The one-standard-error rule chooses the simplest model whose estimated error is within one standard error of the minimum, favoring stability when differences are mostly noise.

0.6.2 Bootstrap and Statistical Uncertainty

The nonparametric bootstrap samples n training observations with replacement, refits the complete procedure, and approximates sampling variability from the replicates. About $1 - e^{-1} \approx 0.632$ of distinct observations appear in a large bootstrap sample; the rest are out-of-bag observations.

Bootstrap standard errors estimate variability of parameters or predictions:

$$\widehat{\text{se}}_{\text{boot}}(\hat{\theta}) = \left\{ \frac{1}{B-1} \sum_{b=1}^B (\hat{\theta}^{*(b)} - \bar{\theta}^*)^2 \right\}^{1/2}.$$

Percentile, basic, and BCa intervals make different corrections. The ordinary bootstrap can fail for nonsmooth statistics, parameters on boundaries, extreme tails, or dependent observations. Use block bootstrap for suitable dependent sequences.

CV fold errors are correlated because fitted training sets overlap; treating them as independent observations understates uncertainty. Comparing two models by paired out-of-fold losses is better than comparing separate averages, but repeated or nested resampling is needed for honest selection uncertainty.

0.7 Model Selection

Model selection chooses features, transformations, hyperparameters, or an algorithm. Training fit always improves as a nested model grows, so selection must penalize complexity or estimate out-of-sample error.

For Gaussian linear models, common criteria are

$$\text{AIC} = -2\ell(\hat{\theta}) + 2d, \quad \text{BIC} = -2\ell(\hat{\theta}) + d \log n,$$

where d is the number of fitted parameters. AIC targets predictive Kullback–Leibler risk; BIC more strongly favors small models and, under restrictive conditions, consistently selects a true finite-dimensional model. They answer different questions.

Searching many models creates selection bias. Standard errors and p -values computed as if the selected model had been fixed in advance are generally too optimistic.

0.7.1 Complexity and Degree of Freedom

For a linear smoother $\hat{y} = Sy$, effective degrees of freedom are

$$\text{df} = \text{tr}(S).$$

OLS has $S = H$ and $\text{df} = p$. For nonlinear procedures, a covariance definition is

$$\text{df} = \frac{1}{\sigma^2} \sum_{i=1}^n \text{Cov}(\hat{y}_i, y_i).$$

It measures sensitivity of fitted values to observed responses, not simply the number of nonzero parameters.

Mallows' C_p and Stein's unbiased risk estimate correct training error by an optimism term proportional to degrees of freedom. Complexity can also be controlled through norm constraints, tree depth, early stopping, margin, or algorithmic stability.

0.8 Ridge Regression

Ridge regression solves

$$\hat{\beta}_\lambda = \arg \min_{\beta} \{\|y - X\beta\|^2 + \lambda \|\beta\|^2\}, \quad \lambda \geq 0, \quad (0.8.1)$$

giving

$$\hat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y.$$

The intercept is normally left unpenalized. Predictors must be standardized because the penalty depends on units. Ridge is also the posterior mode and mean under a Gaussian prior in the Gaussian linear model.

0.8.1 Shrinkage

Conditional on X under $y = X\beta + \varepsilon$,

$$\mathbb{E}[\hat{\beta}_\lambda] = (X^\top X + \lambda I)^{-1} X^\top X \beta,$$

so ridge is biased. It can nevertheless lower prediction MSE by greatly reducing variance, especially under collinearity or $p \geq n$. Unlike lasso, ridge usually retains every feature.

As λ increases, coefficient norm and effective degrees of freedom decrease. Tune λ by validation, fitting standardization within each fold. The constrained form $\|\beta\|^2 \leq t$ is equivalent for a matching Lagrange multiplier.

0.8.2 SVD

Let the centered design have thin SVD $X = UDV^\top$, with singular values $d_1 \geq \dots \geq d_r > 0$. Then

$$\hat{\beta}_\lambda = V \operatorname{diag} \left\{ \frac{d_j}{d_j^2 + \lambda} \right\} U^\top y, \quad \hat{y}_\lambda = U \operatorname{diag} \left\{ \frac{d_j^2}{d_j^2 + \lambda} \right\} U^\top y.$$

Ridge shrinks low-variance directions most strongly. Its effective degrees of freedom are

$$\operatorname{df}(\lambda) = \sum_{j=1}^r \frac{d_j^2}{d_j^2 + \lambda}.$$

The SVD expression is central in interviews: it explains stability, shrinkage, and why ridge remains well-defined when $X^\top X$ is singular.

0.9 Lasso Regression

The lasso solves

$$\hat{\beta}_\lambda = \arg \min_{\beta} \left\{ \frac{1}{2} \|y - X\beta\|^2 + \lambda \|\beta\|_1 \right\}. \quad (0.9.1)$$

It is convex but nonsmooth at zero. The ℓ_1 constraint has corners, so solutions often lie on coordinate axes and are sparse.

0.9.1 Sparsity

Lasso performs shrinkage and variable selection together. With correlated predictors it may select one arbitrarily, making the selected set unstable. Ridge tends to distribute weight across the group. Elastic net combines ℓ_1 and squared ℓ_2 penalties and often stabilizes grouped features.

Exact support recovery needs conditions stronger than those needed for good prediction, including sufficient signal and restrictions on correlations. A sparse fitted model should not automatically be interpreted as identifying the true causal variables.

0.9.2 KKT

Let $r = y - X\hat{\beta}$. The Karush–Kuhn–Tucker conditions are

$$X_j^\top r = \begin{cases} \lambda \operatorname{sign}(\hat{\beta}_j), & \hat{\beta}_j \neq 0, \\ z_j, & |z_j| \leq \lambda, \quad \hat{\beta}_j = 0. \end{cases} \quad (0.9.2)$$

Thus a variable stays at zero when its correlation with the current residual does not exceed λ . These conditions certify optimality and support screening rules.

If $\lambda \geq \|X^\top y\|_\infty$, the all-zero vector is optimal when the response and columns have been centered.

0.9.3 Coordinate Descent

Holding all other coefficients fixed, define the partial residual $r_j = y - \sum_{k \neq j} X_k \beta_k$. The coordinate update is

$$\beta_j \leftarrow \frac{S(X_j^\top r_j, \lambda)}{\|X_j\|^2}, \quad S(z, \lambda) = \text{sign}(z)(|z| - \lambda)_+. \quad (0.9.3)$$

The soft-thresholding operator sets small correlations exactly to zero. Cyclic coordinate descent is efficient for sparse matrices. Warm starts along a decreasing grid of λ values greatly reduce computation; convergence should be checked through KKT residuals or the primal–dual gap.

0.10 Regression Splines and Basis Expansions

A degree- d regression spline is piecewise polynomial with continuity constraints at knots $\xi_1, \dots, \xi_K$. A truncated-power basis is

$$1, x, \dots, x^d, (x - \xi_1)_+^d, \dots, (x - \xi_K)_+^d.$$

Cubic splines are twice continuously differentiable. B-spline bases span the same space but have local support and much better numerical conditioning.

With a fixed basis matrix B , fitting is linear regression. Flexibility is controlled by degree, knot locations, number of knots, or a roughness penalty. Boundary behavior matters: natural cubic splines constrain the function to be linear beyond the boundary knots and often extrapolate more sensibly.

0.10.1 Smoothing Splines

The cubic smoothing spline minimizes

$$\sum_{i=1}^n \{y_i - f(x_i)\}^2 + \lambda \int \{f''(t)\}^2 dt \quad (0.10.1)$$

over functions with square-integrable second derivative. The minimizer is a natural cubic spline with knots at the distinct observed x_i values. Thus an apparently infinite-dimensional optimization has a finite-dimensional solution.

For a suitable basis, the coefficient problem is

$$\hat{\beta} = (B^\top B + \lambda \Omega)^{-1} B^\top y, \quad \Omega_{jk} = \int B_j''(t) B_k''(t) dt.$$

The fitted values are $S_\lambda y$ and $\text{df}(\lambda) = \text{tr}(S_\lambda)$. As $\lambda \rightarrow 0$ the fit interpolates; as $\lambda \rightarrow \infty$ it approaches a straight line. Generalized cross-validation approximates leave-one-out error by

$$\text{GCV}(\lambda) = \frac{n^{-1} \|(I - S_\lambda)y\|^2}{\{1 - \text{tr}(S_\lambda)/n\}^2}.$$

0.10.2 GAMs

A generalized additive model (GAM) assumes

$$g\{\mathbb{E}(Y \mid X = x)\} = \beta_0 + \sum_{j=1}^p f_j(x_j),$$

where g is a link function. Each component is represented by a basis and regularized with a smoothness penalty. Identifiability commonly uses $\sum_i f_j(x_{ij}) = 0$.

GAMs capture nonlinear main effects while retaining component plots and efficient fitting. Backfitting alternately updates each f_j using partial residuals. Penalized IRLS handles non-Gaussian responses. The additive assumption excludes interactions unless terms such as tensor-product smooths are explicitly included.

0.11 Decision Trees

A regression tree partitions feature space into terminal regions $R_1, \dots, R_M$ and predicts

$$\hat{f}(x) = \sum_{m=1}^M \hat{c}_m \mathbb{1}\{x \in R_m\}, \quad \hat{c}_m = \frac{1}{|R_m|} \sum_{i: x_i \in R_m} y_i.$$

At each node, CART greedily chooses a feature j and threshold s minimizing the sum of squared errors in the two child nodes. For classification, common node impurities are

$$Q_{\text{Gini}} = 1 - \sum_k \hat{p}_k^2, \quad Q_{\text{entropy}} = - \sum_k \hat{p}_k \log \hat{p}_k.$$

Trees model interactions automatically, need little preprocessing, and are invariant to monotone transformations of individual features. A single tree is unstable: small data changes can alter early splits and the whole partition. Greedy training is computationally feasible but does not find the globally optimal tree.

Continuous split search can be done after sorting each feature. Missing values require an explicit strategy, such as imputation, learned default directions, or surrogate splits. Unrestricted high-cardinality categorical features can receive an unfairly large chance to reduce training impurity.

0.11.1 Pruning

Grow a sufficiently large tree T_0 , then choose a subtree by cost-complexity pruning:

$$C_\alpha(T) = \sum_{m=1}^{|T|} \sum_{i: x_i \in R_m} (y_i - \hat{c}_m)^2 + \alpha |T|. \quad (0.11.1)$$

Weakest-link pruning produces a nested sequence of optimal subtrees as α increases. Cross-validation chooses α ; the one-standard-error rule often yields a smaller, more stable tree.

Stopping growth early can miss a useful split whose benefit appears only after another split. Growing then pruning separates search from complexity selection.

0.12 Random Forests

Random forests average many deep randomized trees:

$$\hat{f}_{\text{RF}}(x) = \frac{1}{B} \sum_{b=1}^B \hat{f}_b(x).$$

Each tree uses a bootstrap sample, and each split considers a random subset of features. Feature subsampling decorrelates trees; averaging then reduces variance more effectively.

If identically distributed tree predictions have variance σ^2 and pairwise correlation ρ , the average of B trees has variance

$$\rho\sigma^2 + \frac{1-\rho}{B}\sigma^2.$$

Increasing B removes only the second term. Forest performance therefore depends on both individual tree strength and inter-tree correlation.

Out-of-bag predictions evaluate observation i using only trees whose bootstrap samples excluded it. They provide a convenient internal error estimate, though an external test set is still needed after extensive tuning.

0.12.1 Bagging

Bagging fits the same unstable learner to bootstrap samples and averages:

$$\hat{f}_{\text{bag}}(x) = \frac{1}{B} \sum_{b=1}^B \hat{f}^{*(b)}(x).$$

For squared loss, averaging reduces variance. It helps trees because they are unstable, but changes little for an already stable linear estimator.

Permutation importance measures the loss increase after shuffling a feature in out-of-bag data. It measures predictive reliance, not causal effect, and can split importance among correlated variables. Impurity-based importance is biased toward continuous or high-cardinality features.

0.13 Boosting as Functional Gradient Descent

Boosting builds an additive model

$$F_M(x) = F_0(x) + \sum_{m=1}^M \nu \gamma_m h_m(x)$$

by sequentially fitting weak learners. For differentiable empirical loss $\sum_i L(y_i, F(x_i))$, iteration m computes pseudo-residuals

$$r_{im} = - \left. \frac{\partial L(y_i, F)}{\partial F} \right|_{F=F_{m-1}(x_i)},$$

fits h_m to (x_i, r_{im}) , chooses a step γ_m , and updates $F_m = F_{m-1} + \nu\gamma_m h_m$. Thus boosting is gradient descent in function space.

For squared loss, pseudo-residuals are ordinary residuals. For binary logistic loss they are closely related to $y_i - \hat{p}_i$. Tree depth controls interaction order; learning rate ν and number of trees trade off. Smaller ν usually requires more iterations and often generalizes better.

Boosting can overfit, especially with deep trees, noisy labels, or excessive iterations. Tune on validation data and monitor training and validation loss. Unlike bagging, trees are dependent and sequential: each corrects errors of the current ensemble.

Algorithm 1 Gradient boosting

- 1: Initialize $F_0(x) = \arg \min_c \sum_i L(y_i, c)$.
 - 2: **for** $m = 1, \dots, M$ **do**
 - 3: Compute $r_{im} = -\partial L(y_i, F)/\partial F$ at $F = F_{m-1}(x_i)$.
 - 4: Fit a weak learner h_m to $\{(x_i, r_{im})\}$.
 - 5: Choose γ_m by line search.
 - 6: Set $F_m = F_{m-1} + \nu\gamma_m h_m$.
 - 7: **end for**
-

0.14 Maximum Margin Classifiers

For labels $y_i \in \{-1, 1\}$, a linear classifier predicts $\text{sign}(\beta_0 + x^\top \beta)$. For separable data, the hard-margin SVM minimizes $\|\beta\|^2/2$ subject to $y_i(\beta_0 + x_i^\top \beta) \geq 1$. Its geometric margin is $1/\|\beta\|$. Feature scaling matters because distance and the penalty depend on units.

0.14.1 Soft-Margin SVM

Nonseparable data use slack variables:

$$\min_{\beta_0, \beta, \xi} \frac{1}{2} \|\beta\|^2 + C \sum_i \xi_i \quad \text{subject to} \quad y_i(\beta_0 + x_i^\top \beta) \geq 1 - \xi_i, \quad \xi_i \geq 0. \quad (0.14.1)$$

Equivalently, minimize $\|\beta\|^2/2 + C \sum_i [1 - y_i(\beta_0 + x_i^\top \beta)]_+$. Large C gives less regularization; small C permits violations for a wider margin.

0.14.2 SVM Duality

Eliminating the primal variables gives

$$\max_{\alpha} \left\{ \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j x_i^\top x_j \right\} \quad \text{s.t.} \quad 0 \leq \alpha_i \leq C, \quad \sum_i \alpha_i y_i = 0. \quad (0.14.2)$$

The primal coefficient is $\hat{\beta} = \sum_i \hat{\alpha}_i y_i x_i$. Convexity and feasibility give strong duality.

0.14.3 Support Vectors and Kernels

KKT complementary slackness shows that observations with $\alpha_i = 0$ do not directly affect the boundary. Those with $\alpha_i > 0$ are support vectors: $0 < \alpha_i < C$ indicates a point on the margin, while $\alpha_i = C$ indicates a point inside the margin or misclassified.

A positive-semidefinite kernel $K(x, z) = \langle \phi(x), \phi(z) \rangle$ implicitly maps inputs to a feature space. Replacing inner products in (0.14.2) gives

$$f(x) = \hat{\beta}_0 + \sum_i \hat{\alpha}_i y_i K(x_i, x).$$

Polynomial and Gaussian radial kernels are common. Kernel width and C must be tuned jointly. The Gram matrix costs $O(n^2)$ memory. The raw margin is not a probability; calibrate it on held-out data if probabilities are required.

0.15 PCA from Constrained Optimization

For centered $X \in \mathbb{R}^{n \times p}$, the first principal loading solves

$$v_1 = \arg \max_{\|v\|=1} \text{Var}(Xv) = \arg \max_{\|v\|=1} v^\top X^\top X v. \quad (0.15.1)$$

The Lagrange equation makes v_1 the leading covariance eigenvector. Subsequent loadings maximize variance subject to orthogonality. If $X = UDV^\top$, scores are UD , loadings are V , and component j explains variance proportional to d_j^2 . Standardize columns when their scales are not comparable. PCA is unsupervised: high variance need not predict Y .

0.15.1 PCA as LoRA

Here LoRA means low-rank approximation. The Eckart–Young–Mirsky theorem states

$$X_k = U_k D_k V_k^\top = \arg \min_{\text{rank}(A) \leq k} \|X - A\|_F^2, \quad \|X - X_k\|_F^2 = \sum_{j>k} d_j^2. \quad (0.15.2)$$

Modern low-rank adaptation uses an update $\Delta W = AB^\top$. It shares the representation idea, but its factors optimize a supervised objective rather than arise from a data SVD.

0.15.2 PCR connection

Principal components regression regresses y on the first k scores:

$$\hat{\beta}_{\text{PCR},k} = V_k D_k^{-1} U_k^\top y.$$

PCR discards low-variance directions; ridge shrinks them continuously. Because PCR selects directions without y , a low-variance component can be highly predictive. Select k by CV rather than explained variance alone.

0.16 K-means

K -means minimizes

$$\min_{C_1, \dots, C_K, \mu_1, \dots, \mu_K} \sum_{k=1}^K \sum_{i \in C_k} \|x_i - \mu_k\|^2. \quad (0.16.1)$$

Lloyd's algorithm alternates nearest-center assignment with updating each center to its cluster mean. The objective decreases, but convergence is only to a local optimum. Use K -means++ and multiple initializations, standardize differently scaled features, and handle empty clusters explicitly.

K -means favors roughly spherical, similarly sized clusters and is sensitive to outliers. Choose K using stability, substantive usefulness, silhouette summaries, or a model-based criterion; no single diagnostic defines the scientifically correct number.

0.17 Hierarchical Clustering

Agglomerative clustering starts with singleton clusters and repeatedly merges the closest pair. Common linkages are

$$\begin{aligned} d_{\text{single}}(A, B) &= \min_{i \in A, j \in B} d(i, j), \\ d_{\text{complete}}(A, B) &= \max_{i \in A, j \in B} d(i, j), \\ d_{\text{average}}(A, B) &= \frac{1}{|A||B|} \sum_{i \in A, j \in B} d(i, j). \end{aligned}$$

Ward linkage chooses the smallest increase in within-cluster sum of squares.

The dendrogram records merge heights; branch order within a merge is not meaningful. Single linkage can recover elongated clusters but chains through nearby points. Complete linkage favors compact clusters. Distance, scaling, and preprocessing define similarity. A dendrogram always displays structure, even for homogeneous noise, so assess stability by perturbation or bootstrap co-clustering and require substantive validation.